Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

2520030098

Gutha Sushma Sree

Section - 3

Query 1: Create the database

```sql
CREATE DATABASE IF NOT EXISTS bookflows_db;
```

Output:

1 20:24:43 CREATE DATABASE IF NOT EXISTS bookflows_db

Query 2: Select the database

```sql
USE bookflows_db;
```

Output:

2 20:26:17 USE bookflows_db

Query 3: Create the books table

```sql
CREATE DATABASE IF NOT EXISTS bookflows_db;
USE bookflows_db;
CREATE TABLE books (
 book_id INT AUTO_INCREMENT PRIMARY KEY,
 title VARCHAR(255) NOT NULL,
 isbn VARCHAR(13) NOT NULL UNIQUE,
 published_year INT,
 CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
SELECT * FROM books;
```

Output:

3 20:28:11 CREATE TABLE books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, i... 0 row(s) affected

| book_id | title | isbn | published_year |
|---|---|---|---|
| NULL | NULL | NULL | NULL |

Query 4: <u>Create the members table:</u>

```sql
CREATE DATABASE IF NOT EXISTS bookflows_db;
USE bookflows_db;
CREATE TABLE books (
 book_id INT AUTO_INCREMENT PRIMARY KEY,
 title VARCHAR(255) NOT NULL,
 isbn VARCHAR(13) NOT NULL UNIQUE,
 published_year INT,
 CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
SELECT * FROM books;
CREATE TABLE members (
 member_id INT AUTO_INCREMENT PRIMARY KEY,
 full_name VARCHAR(100) NOT NULL,
 email VARCHAR(150) NOT NULL UNIQUE
);
SELECT * FROM members;
```

Output:

| | member_id | full_name | email |
|---|---|---|---|
| * | NULL | NULL | NULL |

Query 5 (verification): <u>Inspect the table structures</u>

```sql
DESCRIBE books;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| | Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|---|
| ▶ | book_id | int | NO | PRI | NULL | auto_increment |
| | title | varchar(255) | NO | | NULL | |
| | isbn | varchar(13) | NO | UNI | NULL | |
| | published_year | int | YES | | NULL | |

```sql
DESCRIBE members;
```

Output:

| | Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|---|
| ▶ | member_id | int | NO | PRI | NULL | auto_increment |
| | full_name | varchar(100) | NO | | NULL | |
| | email | varchar(150) | NO | UNI | NULL | |

Query 6: I<u>nsert 3 books</u>

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Output:

| | | | |
|---|---|---|---|
| 10 | 20:42:08 | INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Cle | |

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Gutha Sushma Sree', 'sush@example.com'),
('Gutha Sowmya Sree', 'sow@example.com'),
('Gutha Mokshagna Naidu', 'moksha@example.com');
```

Output:

| | | |
|---|---|---|
| 14 | 20:44:40 | INSERT INTO members (full_name, email) VALUES ('Gutha Sushma Sree', 'sush@example.com'), ('Gutha Sov |

```
SELECT * FROM books;
```

Output:

| book_id | title | isbn | published_year |
|---|---|---|---|
| 1 | The Alchemist | 9780061122415 | 1988 |
| 2 | Clean Code | 9780132350884 | 2008 |
| 3 | Atomic Habits | 9780735211292 | 2018 |
| NULL | NULL | NULL | NULL |

```
SELECT * FROM members;
```

Output:

| member_id | full_name | email |
|---|---|---|
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |

Constraint Tests (These MUST Fail):

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output:

| | | |
|---|---|---|
| 26 | 09:10:01 | INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000) |

Test 2: NULL title (NOT NULL violation)

```sql
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output:

```
27  09:10:59  INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)
```

Test 3: Future publication year (CHECK violation)

```sql
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Output:

```
28  09:11:59  INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)
```

Test 4: Duplicate email (UNIQUE violation):

```sql
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Output:

```
29  09:13:04  INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')
```

-------------X--------------